CircuitValley
Embedded Engineering

DISTISO6

Galvanically Isolated 6-Output Distribution Amplifier — 100 kHz to 190 MHz

The DISTISO6 is a six-output signal distribution amplifier for clock and reference distribution, covering 100 kHz to 190 MHz with a nominal +4 dB of gain. Every output is galvanically isolated by its own signal transformer, so six instruments can be fed from a single 10 MHz house standard or GPSDO without creating ground loops between them. Each 50 Ω BNC output delivers +13 dBm up to 50 MHz — about 6 Vpp into a 1 MΩ input — and the isolated driver stages hold channel-to-channel skew below the 1 fs resolution of a 70 GHz sampling oscilloscope. The AC-coupled 50 Ω input accepts up to +17 dBm, so a 3.3 V CMOS or TTL clock driving 50 Ω can be connected directly. The unit runs from USB-C PD or a DC barrel jack, in an aluminium bench enclosure.

KEY FEATURES

- Six galvanically isolated 50 Ω BNC outputs — one signal transformer per channel
- 100 kHz to 190 MHz bandwidth, +4 dB nominal gain, flat response
- +13 dBm per output to 50 MHz (≈6 Vpp into 1 MΩ); +12 dBm to 100 MHz; +7 dBm to 190 MHz
- Output isolation breaks ground loops between distributed instruments
- 50 Ω AC-coupled input, up to +17 dBm, tolerates 2 V DC offset
- Accepts 3.3 V CMOS / TTL clocks driving 50 Ω directly
- No measurable channel-to-channel phase difference at 1 fs scope resolution; fractional frequency agreement within ≈10 ppt
- Phase noise –125 dBc/Hz at 1 kHz offset (analyzer noise-floor limited)
- Dual power inputs: USB-C PD (12 V / 15 V) or DC barrel jack 12–24 V
- ≈300 mA typical draw; panel power indicator
- Built for 10 MHz reference, GPSDO and rubidium standard distribution
- CE / UKCA marked, RoHS compliant, built in Germany
- Open hardware — design files and documentation published



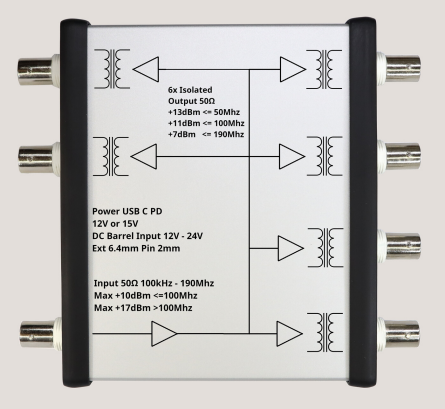
DISTISO6 — six isolated 50 Ω BNC outputs in an aluminium bench enclosure

SPECIFICATIONS

Configuration	One 50 Ω input driving six transformer-isolated 50 Ω outputs
Frequency range	100 kHz to 190 MHz
Gain	+4 dB nominal, flat across the band (see measured response)
Input impedance	50 Ω, AC-coupled to the primary stage
Maximum input level	+10 dBm, 100 kHz–100 MHz (2 Vpp, up to 2 V DC offset) +17 dBm, 101–190 MHz
Input compatibility	3.3 V CMOS or TTL driving into 50 Ω; use an external BNC attenuator for higher drive levels
Outputs	6 × 50 Ω, each galvanically isolated by its own signal transformer
Output level	+13 dBm, 100 kHz–50 MHz (≈6 Vpp into 1 MΩ) +12 dBm, 50–100 MHz • +7 dBm, 101–190 MHz (into 50 Ω)
Phase noise (10 MHz)	–70 dBc/Hz @ 1 Hz • –90 dBc/Hz @ 10 Hz • –114 dBc/Hz @ 100 Hz • –125 dBc/Hz @ 1 kHz — limited by the noise floor of a generic spectrum analyzer; the device performs at least this well
Channel-to-channel phase	No measurable difference at 1 fs resolution (Tektronix TDS8000, 70 GHz bandwidth, 50 GHz and 20 GHz sampling heads)
Channel-to-channel frequency	Same-channel measurement baseline 7.956 ppt; outputs on the same driver ≈8.16 ppt; outputs on different drivers ≈10.2 ppt (≈10 ppt worst case)
Connectors	BNC female 50 Ω: outputs 1–4 front, outputs 5–6 and input on the rear panel
Power input	USB-C Power Delivery, 12 V or 15 V • DC barrel jack 12–24 V, 6.5 mm outer / 2.0 mm centre pin, centre positive
Supply current / indicator	≈300 mA typical; power LED on the rear panel
Enclosure	Extruded aluminium with moulded end caps; benchtop / stackable
Documentation	Design files, source and application notes published at github.com/circuitvalley
Compliance	CE / UKCA marked; RoHS; WEEE registered
Country of origin	Germany

APPLICATIONS & ARCHITECTURE

The DISTISO6 is intended for distributing a single frequency or clock reference — a 10 MHz GPSDO, rubidium or OCXO house standard, a word clock, or a pulse train — to a bench full of instruments. The input is AC-coupled into a 50 Ω primary stage, split, and fed to six independent driver stages, each of which drives its BNC output through its own signal transformer. That transformer is what makes the outputs galvanically isolated: no DC path exists between any two connected instruments, so mains-earthed test gear can be tied to the same reference without forming ground loops, and a fault on one output cannot pull the others with it. Because each channel has its own driver and transformer, loading or shorting one output does not disturb the remaining five. Two power inlets are provided — USB-C PD and a DC barrel jack — so the amplifier can run from a bench supply, a battery, or a laptop-class USB-C adapter.



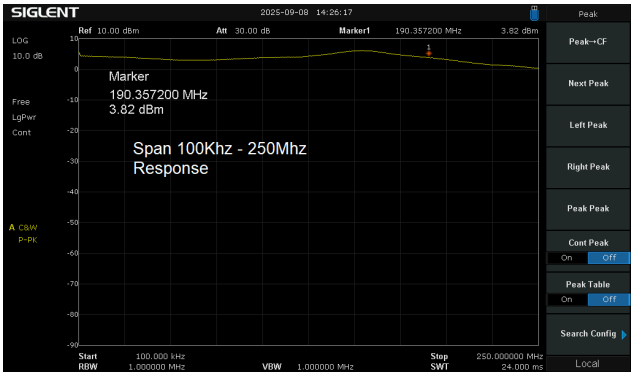
Signal path: one input, six transformer-isolated drivers



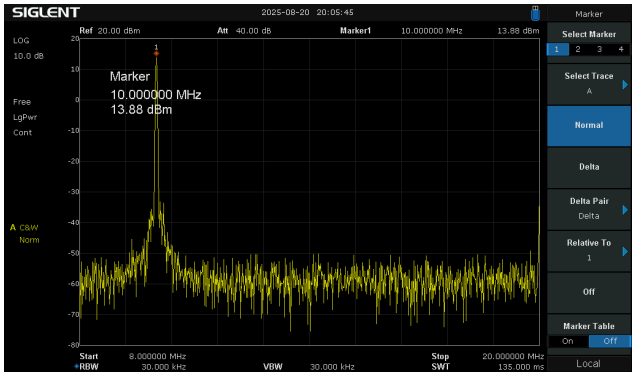
Front panel (outputs 1–4) and rear panel (outputs 5–6, input, USB-C PD and DC power)

MEASURED PERFORMANCE

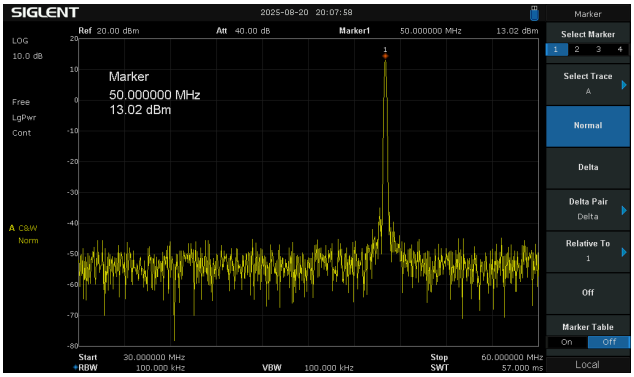
Captures taken on a Siglent SSA spectrum analyzer at a 50 Ω output, driven from a 10 MHz reference source:



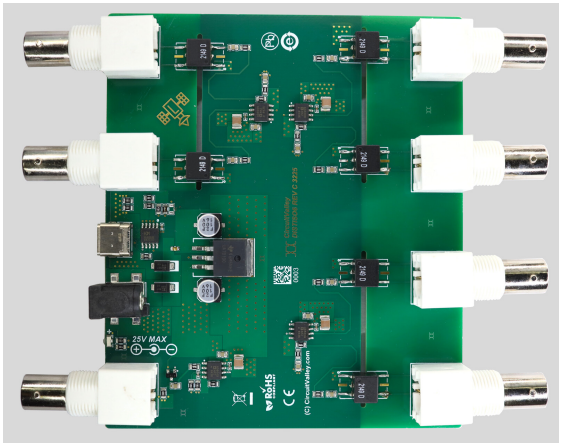
Frequency response, 100 kHz to 250 MHz span — flat in-band, marker 190.357 MHz at 3.82 dBm



10 MHz output level — marker 10.000000 MHz at 13.88 dBm into 50 Ω



50 MHz output level — marker 50.000000 MHz at 13.02 dBm into 50 Ω



Internal assembly — one driver and one isolation transformer per output channel

Gaurav Singh — CircuitValley
Ried 43, 88161 Lindenberg, Germany
E-Mail: gauravsingh@circuitvalley.com

Geschäftsführer: Gaurav Singh | Amtsgericht Lindenberg
USt.-IdNr. DE452805793
WEEE-Reg.-Nr.: DE90905884